EN 580.133
mRNA as Therapeutics (Part II)

Jan 7, 2025

Created by Elaine Zhelan Chen

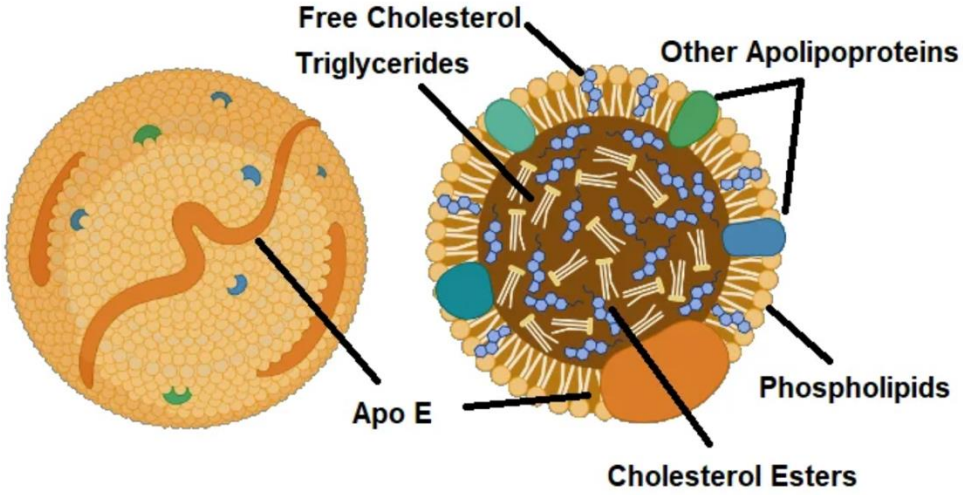
Recap

- RNA therapeutics have the potential to target any genetic components within the cell
- RNA therapeutics can target previously undruggable targets using small molecules and biologics
- mRNA provides the instruction for making proteins on the ribosomes in the cytoplasm
- mRNA requires a delivery system that protects the nucleic acid from degradation, allows cellular uptake, and helps targeting a specific cell type
- LNPs are a class of highly efficient delivery system for exogenous mRNA in vivo

mRNA delivery

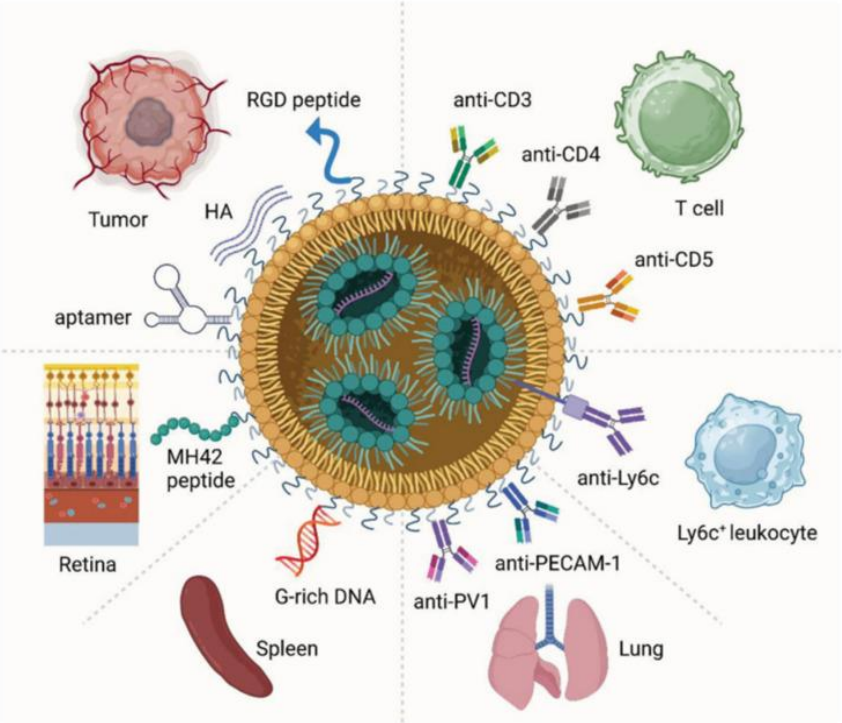
Hepatic delivery

- Binding of ApoE to cholesterol molecules facilitates their absorption by hepatocytes
- Similarly to the hepatic metabolism of low-density lipoprotein (LDL)
- Valuable for liver hepatocyte applications
- Limits the applications of LNP outside of liver hepatocytes – extrahepatic delivery



Targeting specific cell types

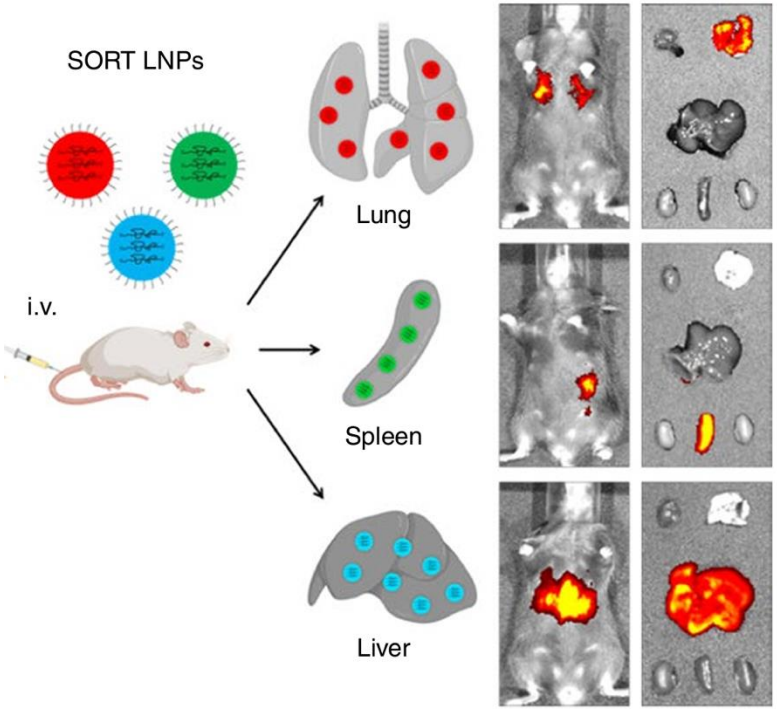
- Cell type tropism
- anti-Ly6c antibody–conjugated LNPs to target Ly6c+ leukocytes
- anti-PECAM1 antibody–conjugated LNPs to target pulmonary endothelium
- anti-CD4 antibody–conjugated LNPs to target T cells
- anti-hCD29 antibody–conjugated LNPs to target lymphoma cells
- anti-CD5 antibody–conjugated LNPs to target CAR-T cells
- Active targeting



Decorating the surface of the LNP with antibodies, peptides, carbohydrates

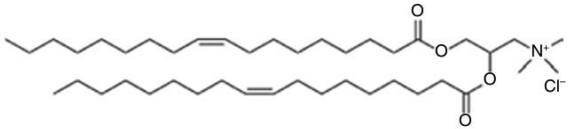
Targeting specific tissue

- Tissue tropism
- Net charge of the LNP affects the tissue of target
  - Positively charged -> lung
  - Neutral particles -> livers
  - Negatively charged -> spleen
- Passive targeting
- SORT (Selective Organ Targeting) Lipid nanoparticles

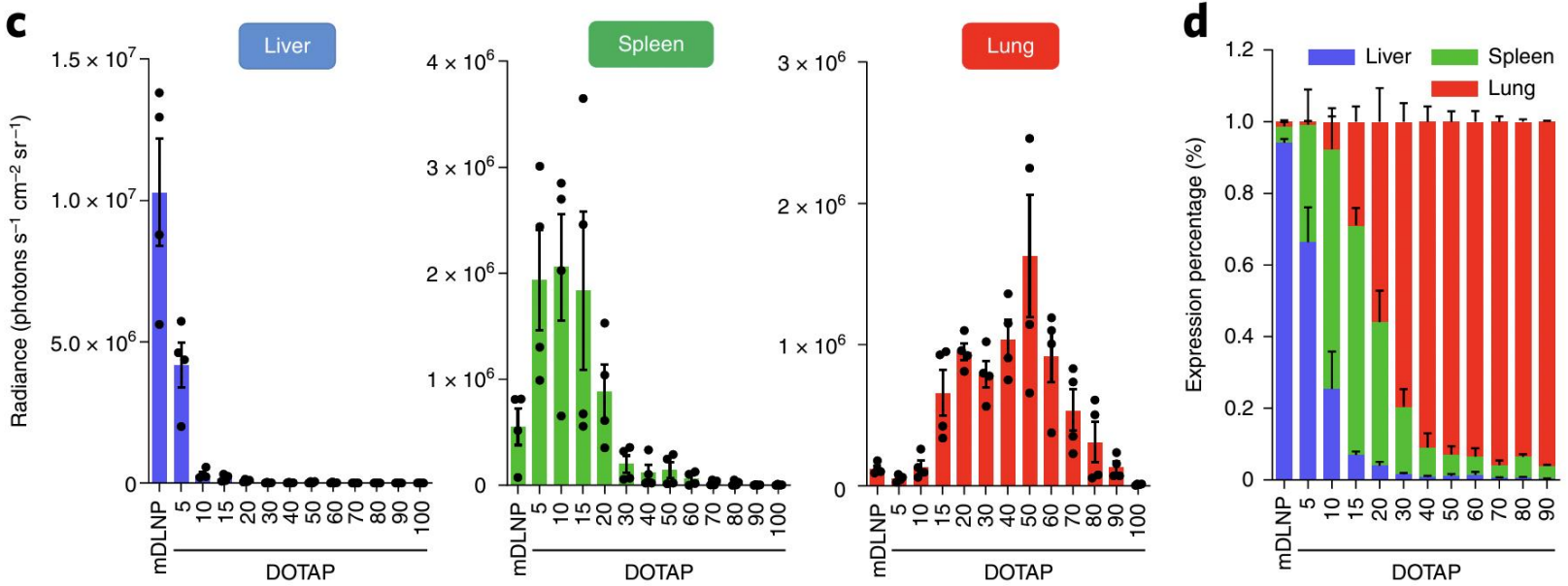


SORT study

DOTAP

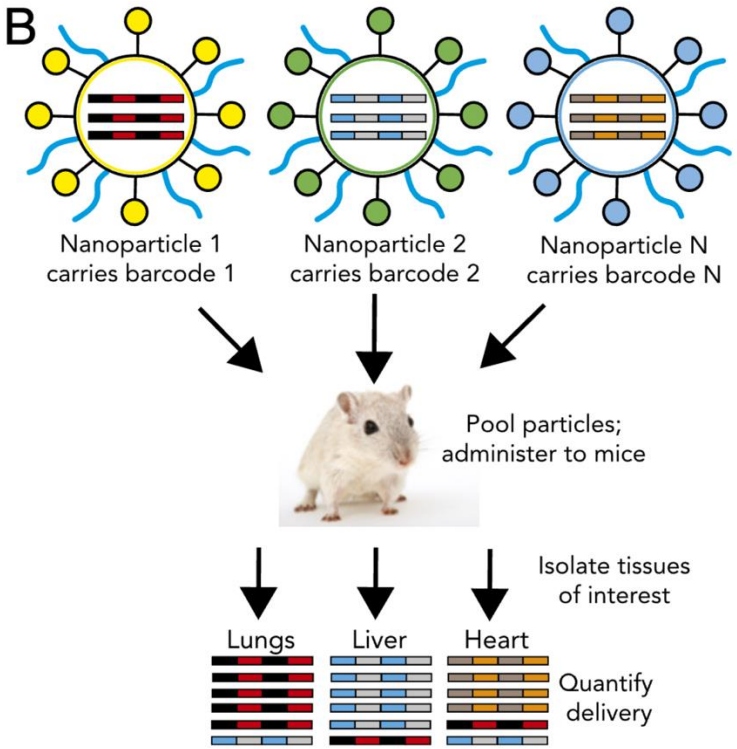


permanently cationic lipid



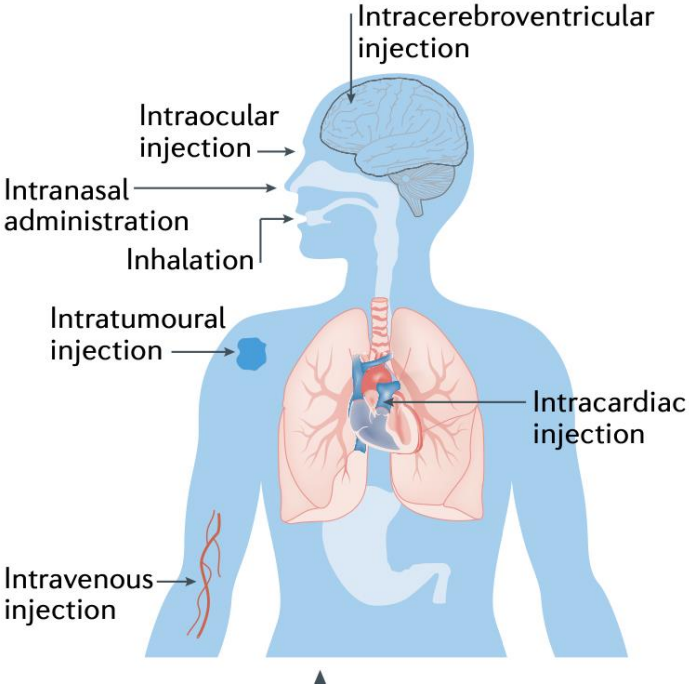
LNP formulation screening

- High throughput
- Quantified particle biodistribution by deep sequencing the barcodes
- measuring the biodistribution of 30 nanoparticles to eight tissues simultaneously



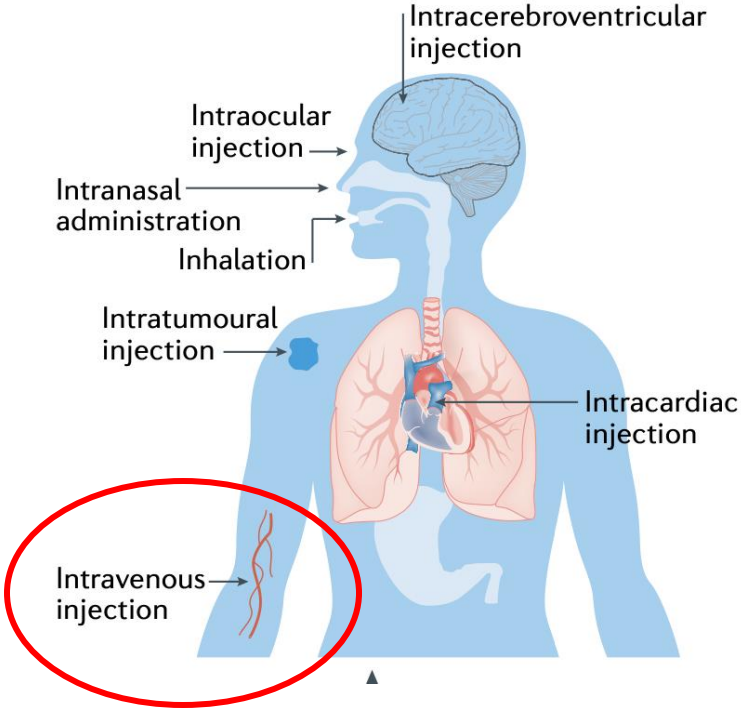
Routes of administration

- Administration routes can greatly influence organ distribution, expression kinetics and therapeutic outcomes of lipid nanoparticle–mRNA formulations
- The administration route is often determined by the properties of the nanoparticles and therapeutic indications



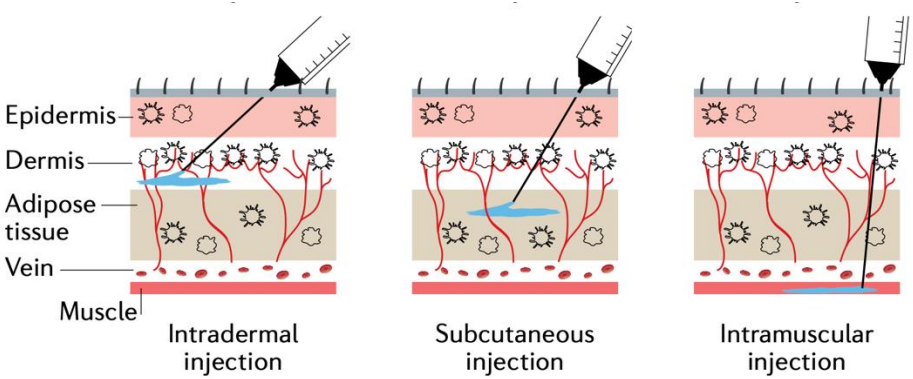
intravenous (i.v.)

- Lipid nanoparticles accumulates in the liver
- The liver is inherently capable of producing secretory proteins
- i.v. administration can be used to produce proteins that are missing or to produce antibodies to neutralize pathogens or target cancer cells
- For extrahepatic delivery, addition targeting strategies need to be deployed



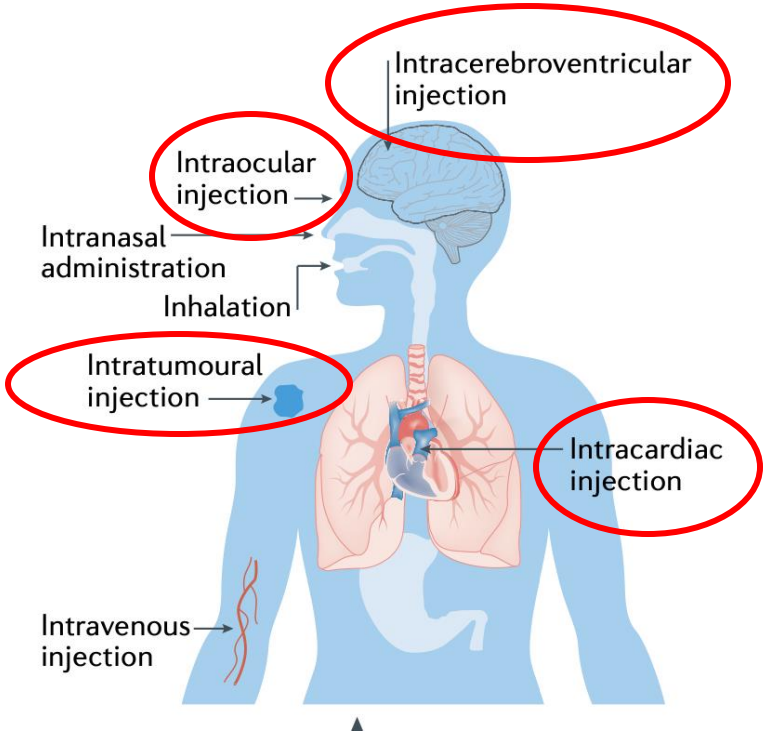
Local administration

- Intradermal (i.d.), intramuscular (i.m.) and subcutaneous (s.c.) injection
- Commonly used for vaccination because resident and recruited antigen-presenting cells (APCs) are present in the skin and muscle
- Nearby APCs can internalize and process mRNA-encoded antigens to stimulate T cell immunity
- Intranasal administration also benefits from the proximity of APCs in the peripheral lymph nodes



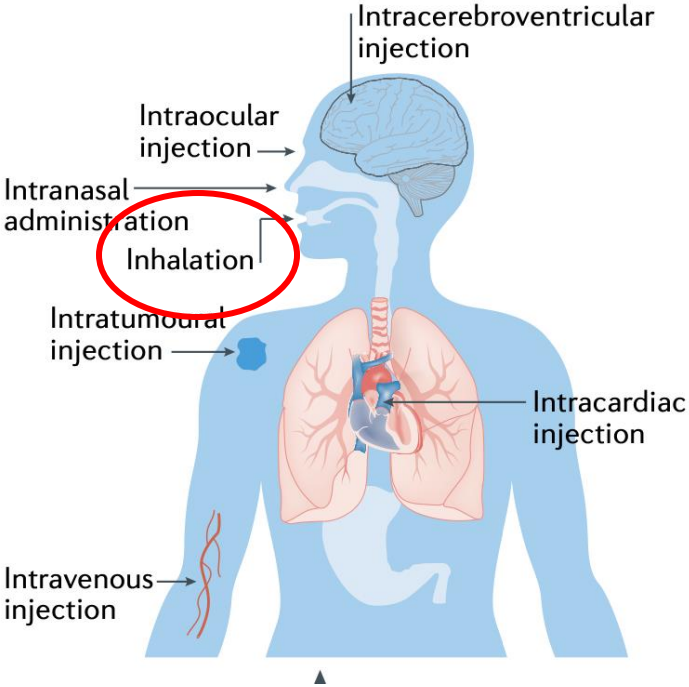
Local administration

- Aims to achieve local therapeutic effects
- Local injection of LNP–mRNA formulations enables supplementation of therapeutic proteins in specific tissues, such as heart, eyes, and brain
- LNP–mRNA formulations can be directly delivered into tumor tissue by intratumoural injection to initiate immune cell activation



Inhalation

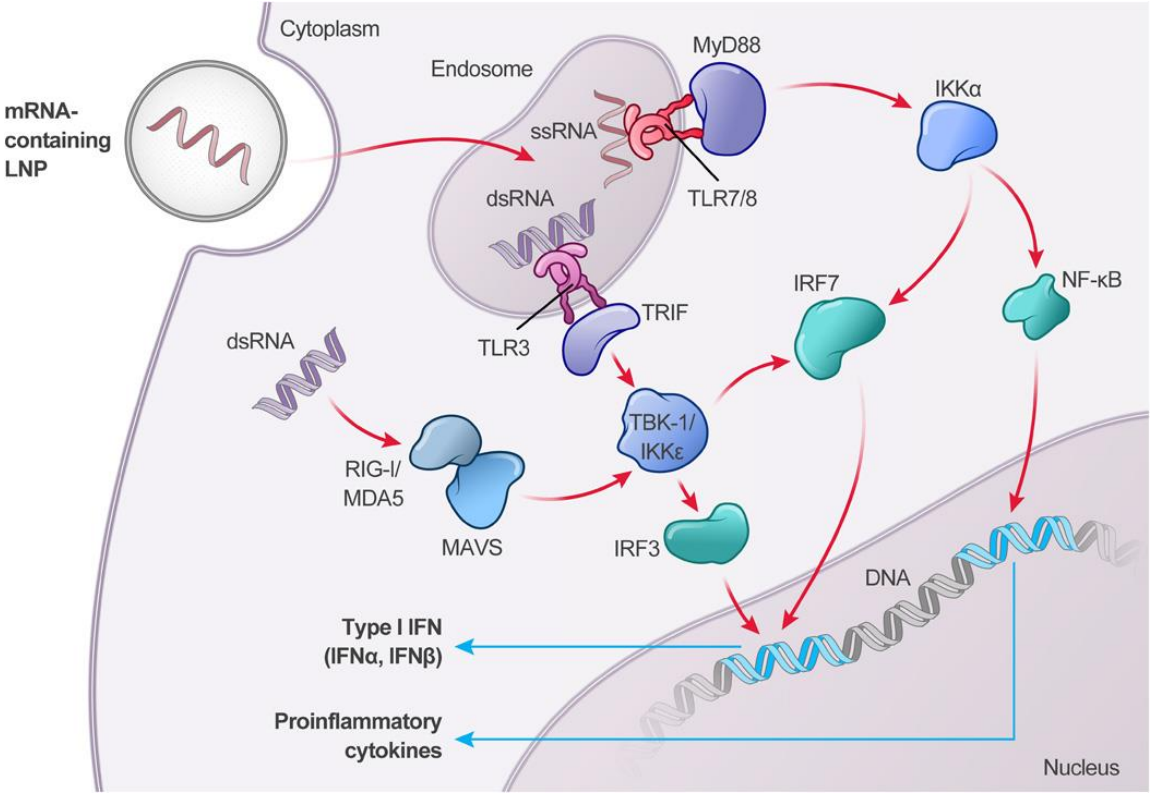
- LNP–mRNA formulations can be administered directly into the lungs by inhalation
- Provides direct, rapid and non-invasive access to the lung
- However, the mRNA must be highly concentrated to withstand the shear forces during aerosolization
- A potential route for non-invasive drug delivery to the brain and the central nervous system



mRNA immunogenicity

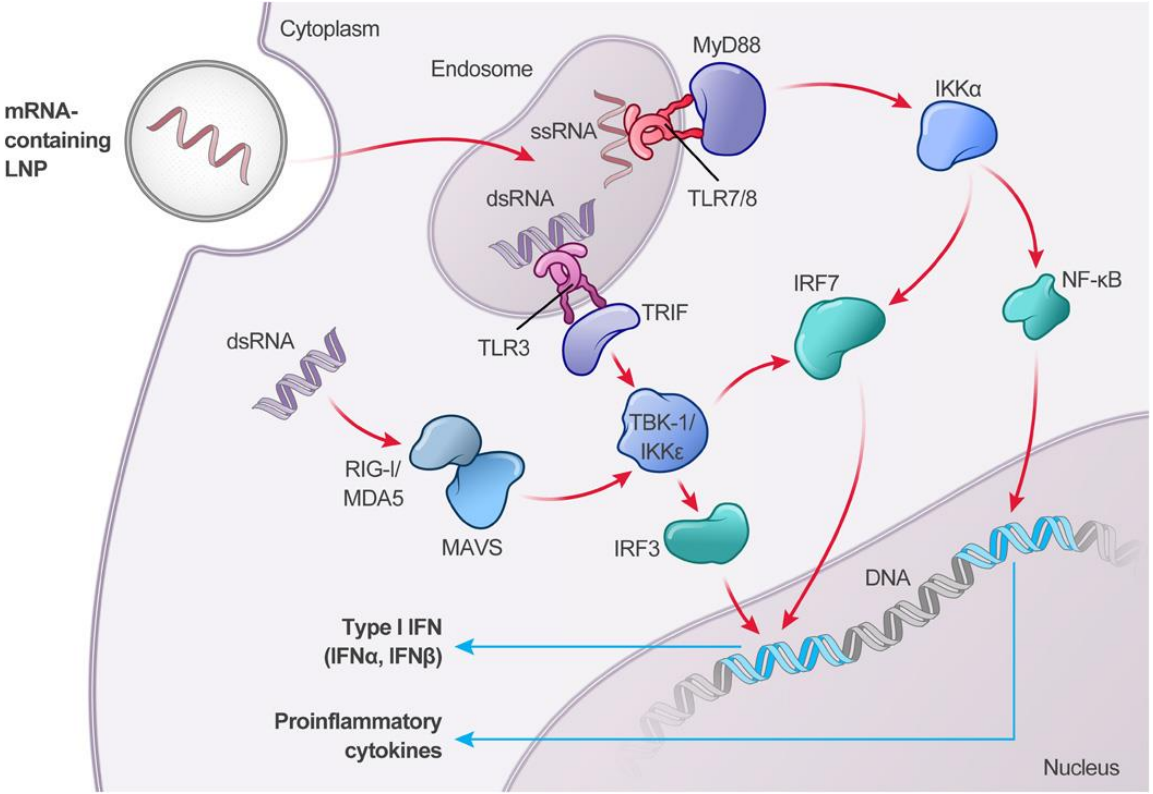
Innate immune response

- Exogenous RNA will be recognized as a signal of viral infection
- RNA stimulate the mammalian innate immune system through activation of Toll-like receptors (TLRs)
- Triggering of TLR activates signal transduction pathways that selectively lead to production of proinflammatory cytokines, which collectively act to eliminate intracellular pathogens and infected cells



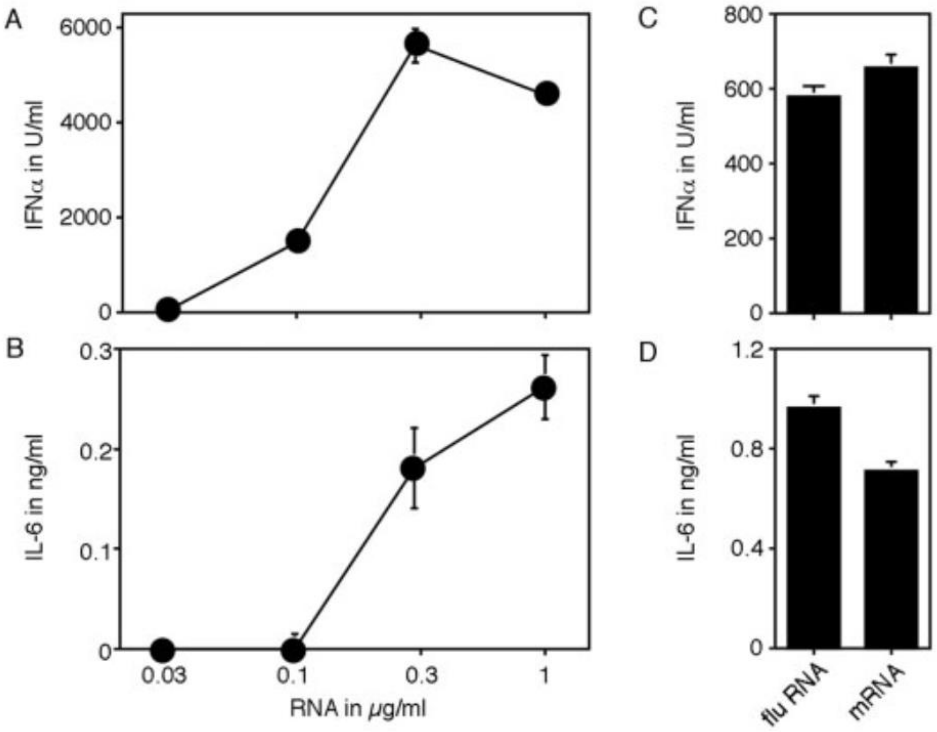
Consequences

- Reduced mRNA translation which decreases the potency of therapeutic mRNA
- Activate stress response genes can ultimately inhibit protein translation
- Detrimental for chronic disease, which needs repeated dosing
- This natural cellular defense mechanism aims to block protein expression of viral genes but has the added consequence of reducing overall cellular gene expression



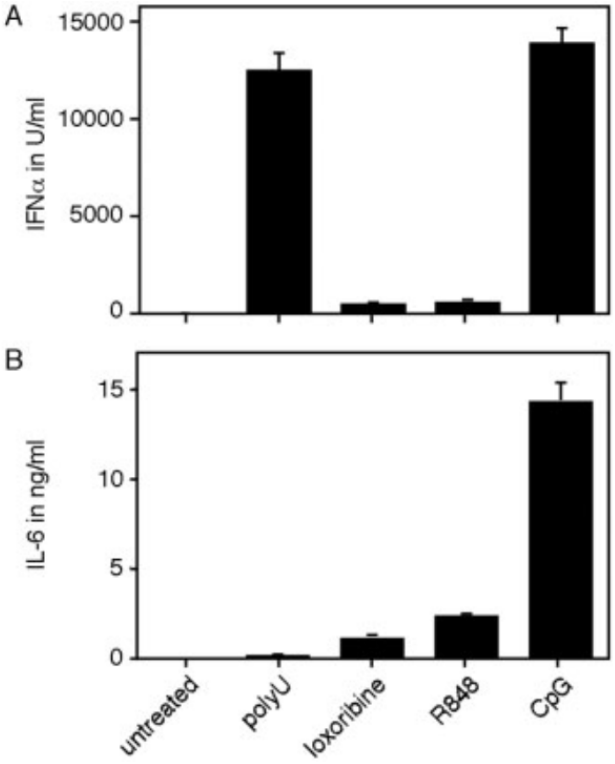
mRNA can activate innate immune response

- Treating mice with flu RNA vs mouse spleen mRNA
- mRNA can act as an innate immune stimulus

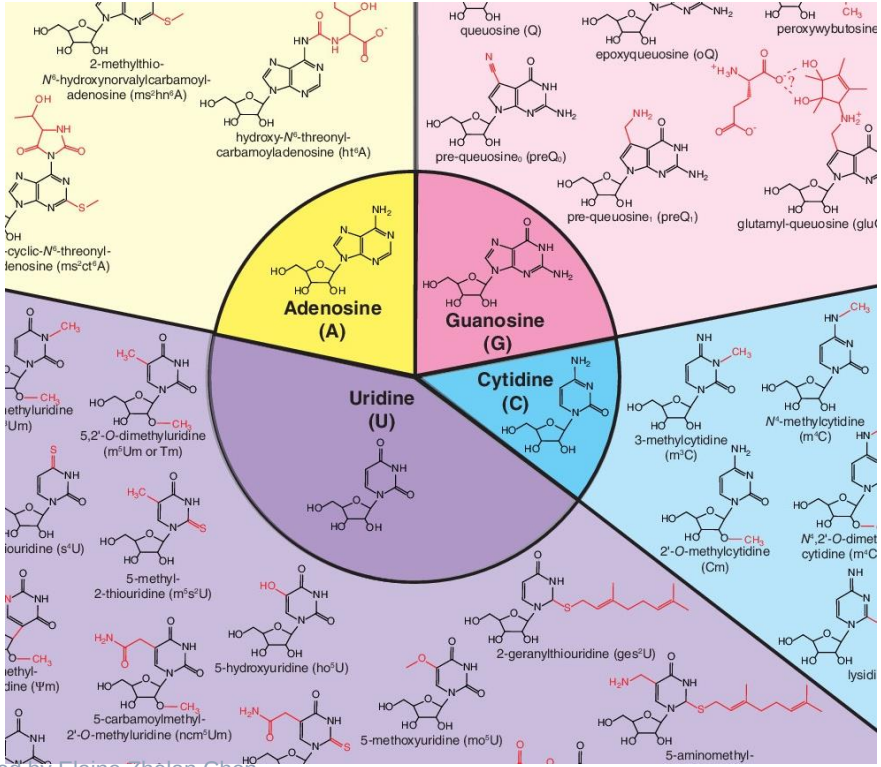


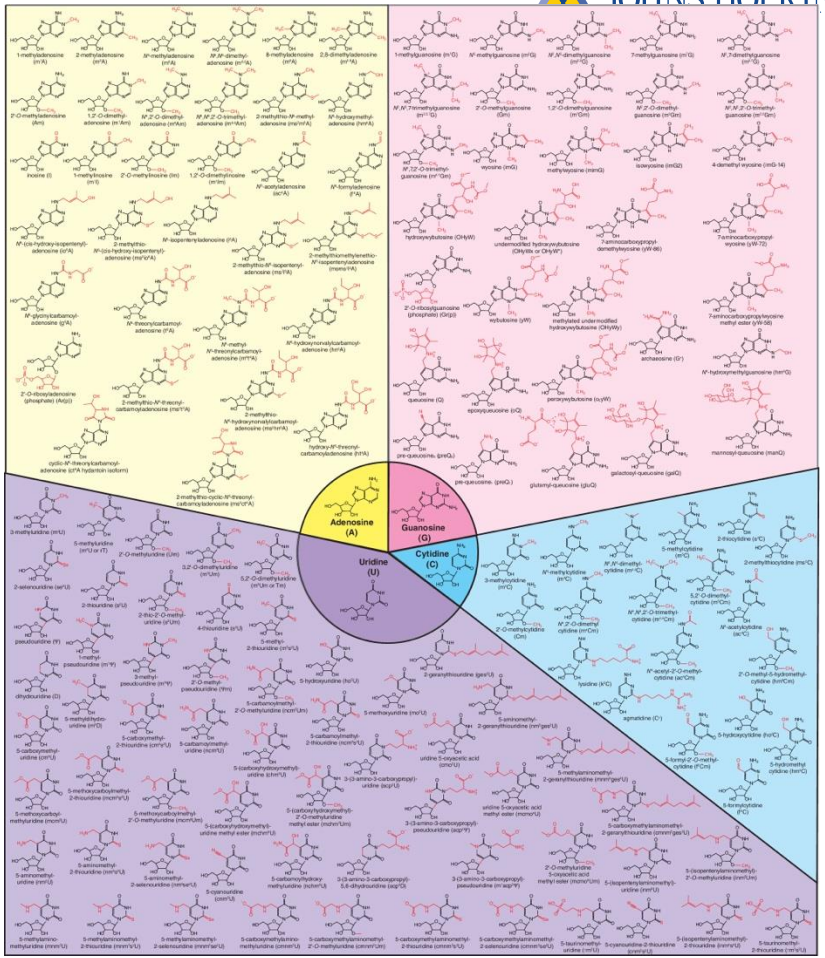
Uridine to be specific

- TLR7 recognition relies on sensing of uracil and ribose, the two distinguishing features of RNA
- Repeats of two or three uridine clusters are sufficient to activate TLR7



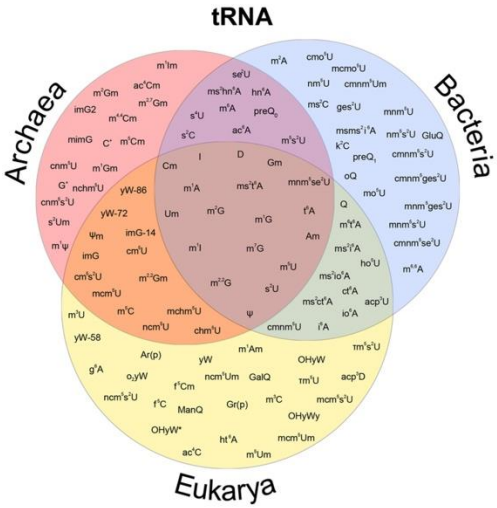
Naturally occurring modified nucleotides

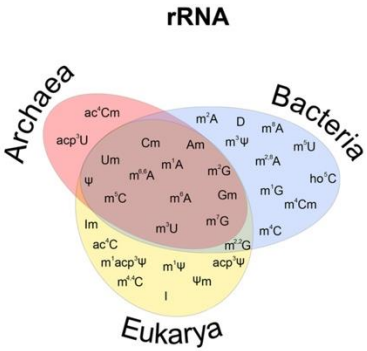


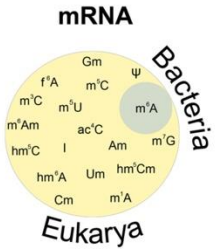


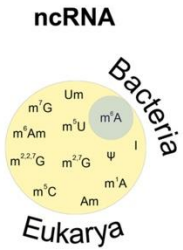
Presence in cells

- Ribosomal RNA (rRNA) and transfer RNA (tRNA) are both heavily modified in human cells
- Together rRNA and tRNA constitute over 80% of cellular RNA
- The role of nucleotide modifications on the immunostimulatory potential of RNA is not known.

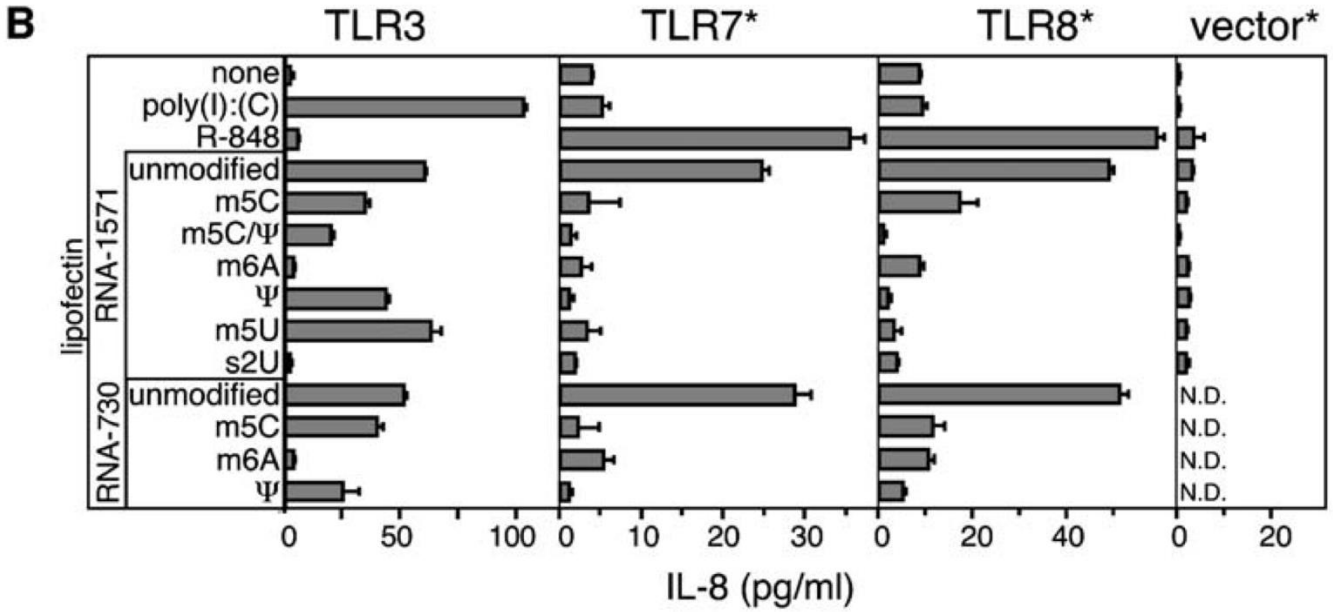






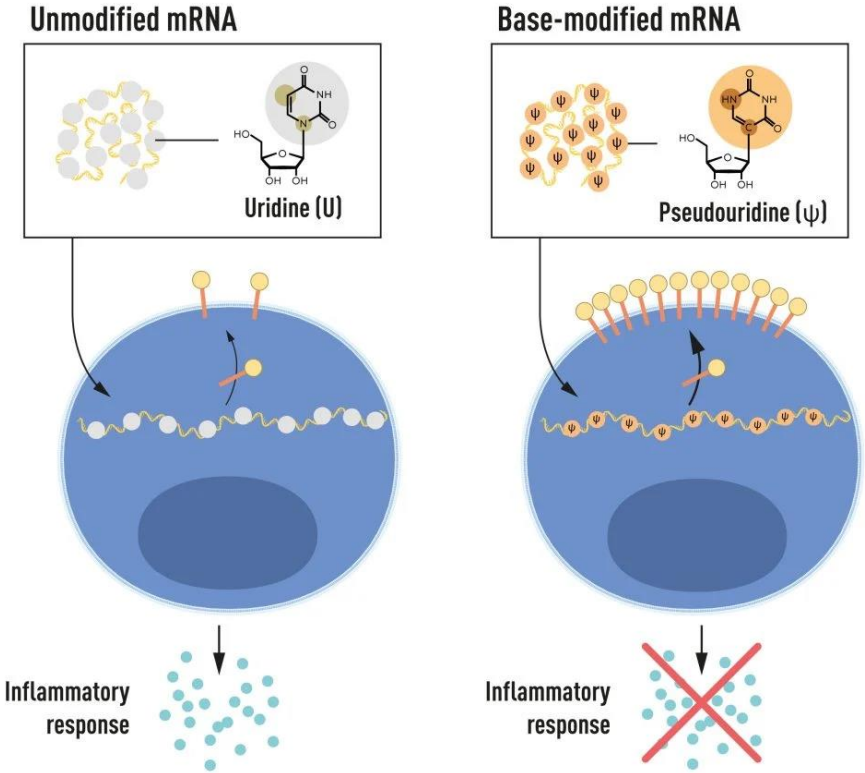


Modified nucleotides



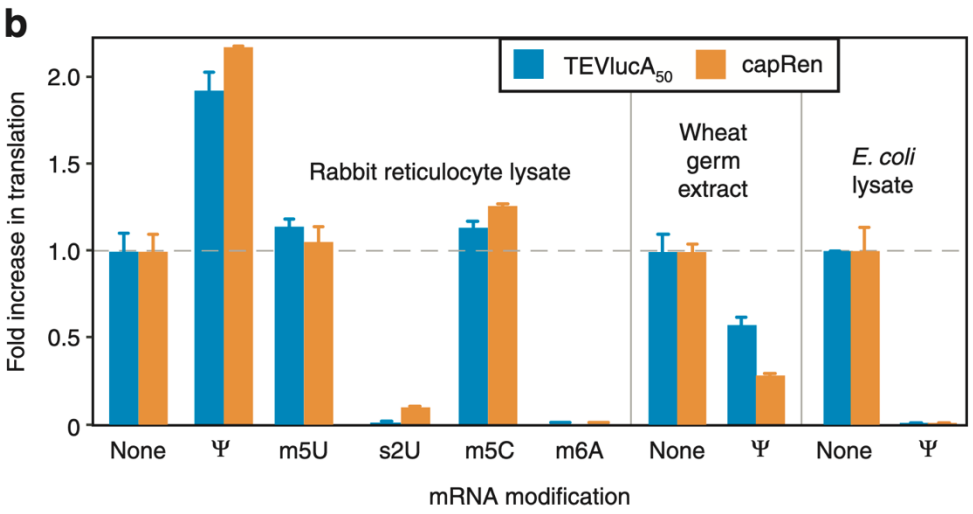
Unmodified RNA stimulates human TLR3, TLR7, and TLR8, but most of the nucleoside-modified RNAs Are Not Stimulatory

Modified nucleotides



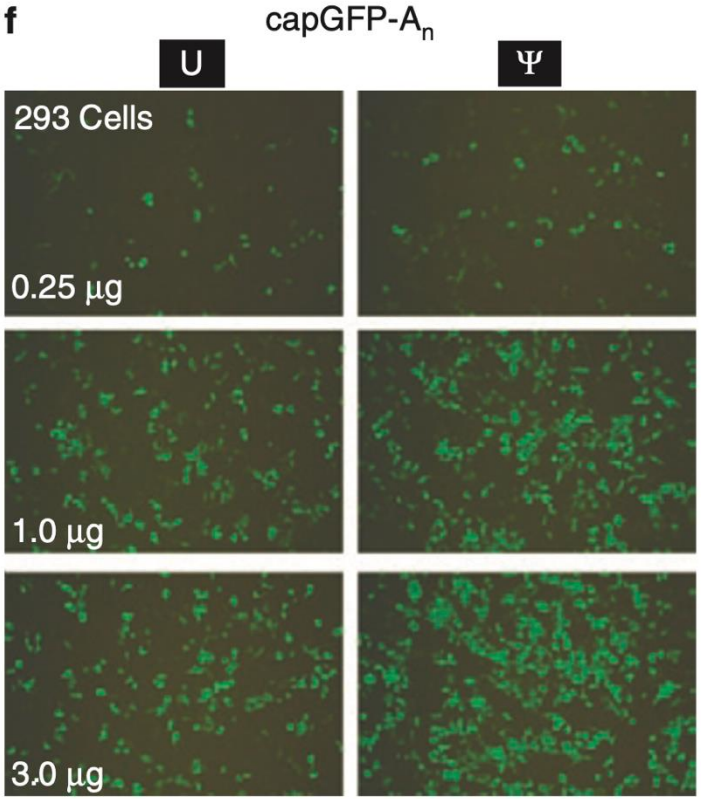
Pseudouridine Ψ

- A naturally modified nucleotide
- mRNAs containing pseudouridines have a higher translational capacity than unmodified mRNAs
- At higher doses, only the unmodified mRNA was immunogenic, inducing high serum levels of interferon-α (IFN-α)
- Enhance stability and translational capacity of mRNA while diminishing its immunogenicity in vivo



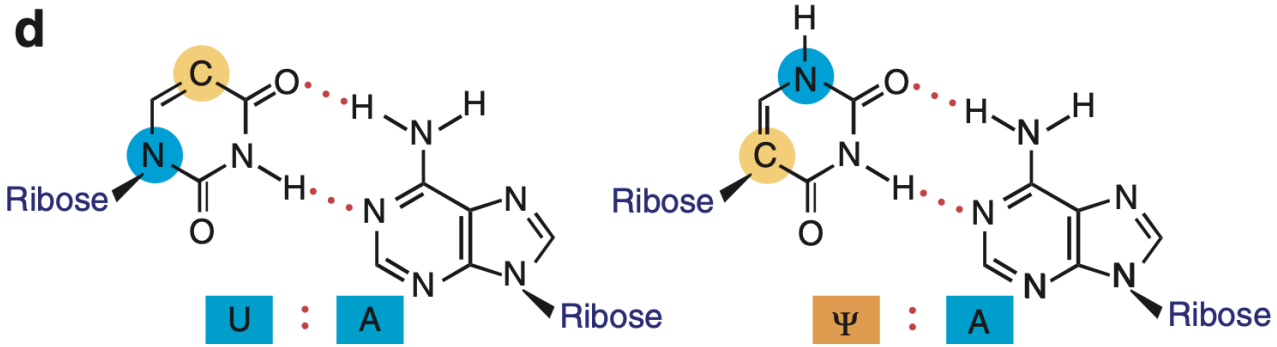
Pseudouridine Ψ

- A higher expression of Green Fluorescent Protein (GFP) under the same mRNA dosage by incorporating pseudouridine



Pseudouridine Ψ

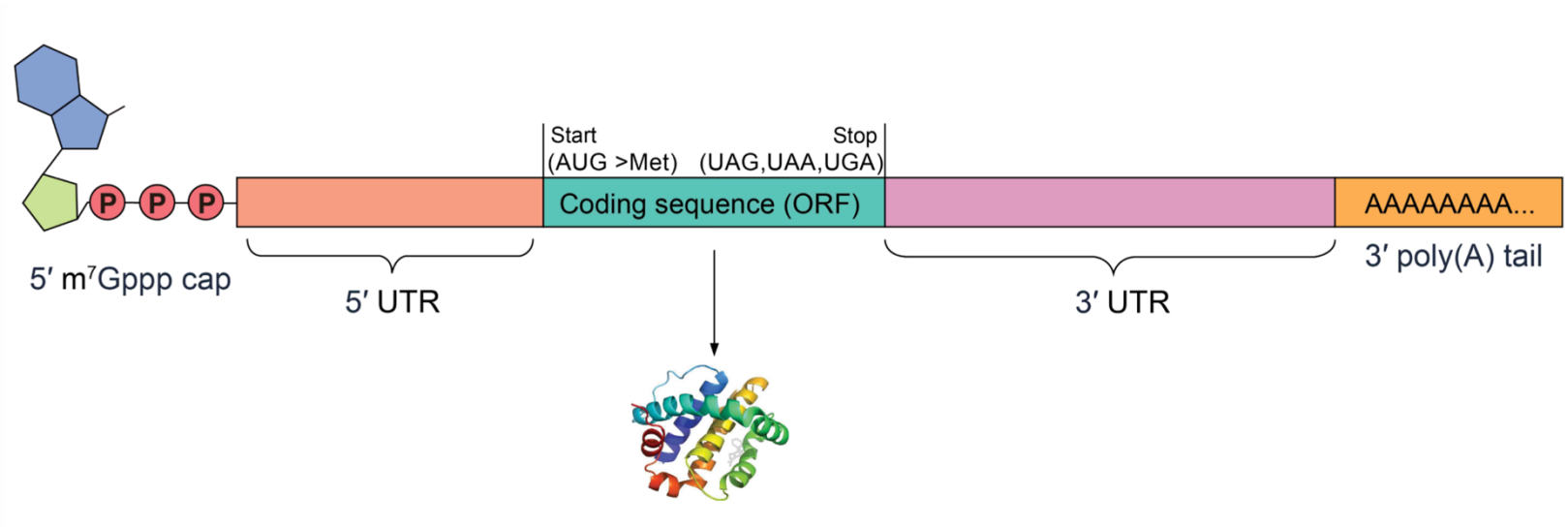
- Ψ has not been found in any mRNA
- However, its ability to form hydrogen bonds with adenosine is similar to that of uridine



Modified nucleotides

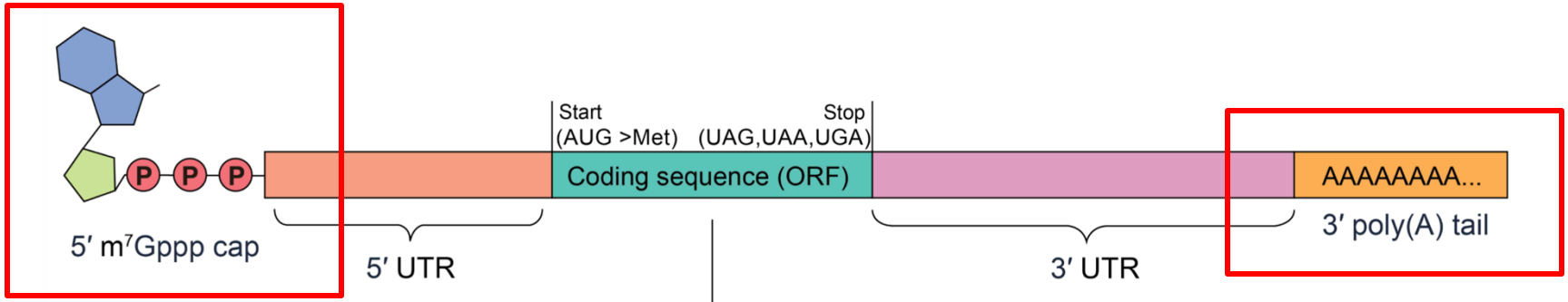
2023 Nobel Prize in Physiology or Medicine
Katalin Karikó and Drew Weissman

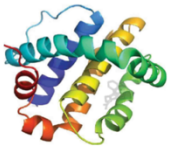
mRNA structure



A mature mRNA, consisting of the 5’ and 3’ untranslated regions (UTRs), an open reading frame (ORF), a cap structure and a poly(A) tail

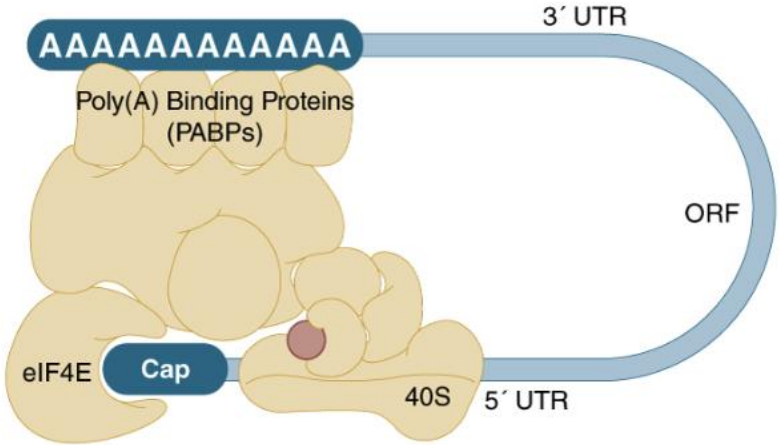
mRNA structure





- Cap protects the mature mRNA from degradation and serves a role in nuclear export and efficient translation.
- Poly(A) tail stabilizes mRNA and aids in the export of mRNA to the cytoplasm

Translation initiation complex



mRNA forms a “closed-loop” structure via interactions mediated by protein complexes that bind the cap structure and the poly(A) tail

mRNA optimization

Purposes

- Minimize innate immune response
- Enhance mRNA stability
- Maximize translation

Sequence/Codon optimization

- Multiple codons can encode for the same amino acid
- Therefore, multiple RNA sequences can encode for the same protein
- mRNA sequence can be optimized in terms of **codon usage** to **enhance protein expression**
- Enhance translation rates by replacing rare codons with synonymous ones favored by the cellular environment
- This is guided by the abundance of transfer RNAs (tRNAs) in host cells

Second letter

First letter	U	C	A	G	Third letter
U	UUU, UUC } Phe UUA, UUG } Leu	UCU, UCC, UCA, UCG } Ser	UAU, UAC } Tyr **UAA Stop** **UAG Stop**	UGU, UGC } Cys **UGA Stop** UGG Trp	U C A G
C	CUU, CUC, CUA, CUG } Leu	CCU, CCC, CCA, CCG } Pro	CAU, CAC } His CAA, CAG } Gln	CGU, CGC, CGA, CGG } Arg	U C A G
A	AUU, AUC, AUA } Ile AUG Met	ACU, ACC, ACA, ACG } Thr	AAU, AAC } Asn AAA, AAG } Lys	AGU, AGC } Ser AGA, AGG } Arg	U C A G
G	GUU, GUC, GUA, GUG } Val	GCU, GCC, GCA, GCG } Ala	GAU, GAC } Asp GAA, GAG } Glu	GGU, GGC, GGA, GGG } Gly	U C A G

Codon usage bias

***Homo sapiens* [gbpri]: 93487 CDS's (40662582 codons)**

fields: [triplet] [frequency: **per thousand**] ([number])

```
UUU 17.6(714298)  UCU 15.2(618711)  UAU 12.2(495699)  UGU 10.6(430311)
UUC 20.3(824692)  UCC 17.7(718892)  UAC 15.3(622407)  UGC 12.6(513028)
UUA  7.7(311881)  UCA 12.2(496448)  UAA  1.0( 40285)  UGA  1.6( 63237)
UUG 12.9(525688)  UCG  4.4(179419)  UAG  0.8( 32109)  UGG 13.2(535595)

CUU 13.2(536515)  CCU 17.5(713233)  CAU 10.9(441711)  CGU  4.5(184609)
CUC 19.6(796638)  CCC 19.8(804620)  CAC 15.1(613713)  CGC 10.4(423516)
CUA  7.2(290751)  CCA 16.9(688038)  CAA 12.3(501911)  CGA  6.2(250760)
CUG 39.6(1611801)  CCG  6.9(281570)  CAG 34.2(1391973)  CGG 11.4(464485)

AUU 16.0(650473)  ACU 13.1(533609)  AAU 17.0(689701)  AGU 12.1(493429)
AUC 20.8(846466)  ACC 18.9(768147)  AAC 19.1(776603)  AGC 19.5(791383)
AUA  7.5(304565)  ACA 15.1(614523)  AAA 24.4(993621)  AGA 12.2(494682)
AUG 22.0(896005)  ACG  6.1(246105)  AAG 31.9(1295568)  AGG 12.0(486463)

GUU 11.0(448607)  GCU 18.4(750096)  GAU 21.8(885429)  GGU 10.8(437126)
GUC 14.5(588138)  GCC 27.7(1127679)  GAC 25.1(1020595)  GGC 22.2(903565)
GUA  7.1(287712)  GCA 15.8(643471)  GAA 29.0(1177632)  GGA 16.5(669873)
GUG 28.1(1143534)  GCG  7.4(299495)  GAG 39.6(1609975)  GGG 16.5(669768)
```

Coding GC 52.27% 1st letter GC 55.72% 2nd letter GC 42.54% 3rd letter GC 58.55%

Codon Adaptation Index (CAI)

- A higher CAI value indicates a greater similarity to the codon usage of the host organism
- Using the most prevalent codons for all amino acids in an ORF results in a CAI of 1.0
- Approved mRNA vaccines have optimized ORFs with CAI values exceeding 0.9
- The sequence has been modified to closely resemble the preferred codon usage of the host organism,
- Enhanced protein expression level

***Homo sapiens* [gbpri]: 93487 CDS's (40662582 codons)**

fields: [triplet] [frequency: **per thousand**] ([number])

```
UUU 17.6(714298)  UCU 15.2(618711)  UAU 12.2(495699)  UGU 10.6(430311)
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UUA  7.7(311881)  UCA 12.2(496448)  UAA  1.0( 40285)  UGA  1.6( 63237)
UUG 12.9(525688)  UCG  4.4(179419)  UAG  0.8( 32109)  UGG 13.2(535595)

CUU 13.2(536515)  CCU 17.5(713233)  CAU 10.9(441711)  CGU  4.5(184609)
CUC 19.6(796638)  CCC 19.8(804620)  CAC 15.1(613713)  CGC 10.4(423516)
CUA  7.2(290751)  CCA 16.9(688038)  CAA 12.3(501911)  CGA  6.2(250760)
CUG 39.6(1611801)  CCG  6.9(281570)  CAG 34.2(1391973)  CGG 11.4(464485)

AUU 16.0(650473)  ACU 13.1(533609)  AAU 17.0(689701)  AGU 12.1(493429)
AUC 20.8(846466)  ACC 18.9(768147)  AAC 19.1(776603)  AGC 19.5(791383)
AUA  7.5(304565)  ACA 15.1(614523)  AAA 24.4(993621)  AGA 12.2(494682)
AUG 22.0(896005)  ACG  6.1(246105)  AAG 31.9(1295568)  AGG 12.0(486463)

GUU 11.0(448607)  GCU 18.4(750096)  GAU 21.8(885429)  GGU 10.8(437126)
GUC 14.5(588138)  GCC 27.7(1127679)  GAC 25.1(1020595)  GGC 22.2(903565)
GUA  7.1(287712)  GCA 15.8(643471)  GAA 29.0(1177632)  GGA 16.5(669873)
GUG 28.1(1143534)  GCG  7.4(299495)  GAG 39.6(1609975)  GGG 16.5(669768)
```
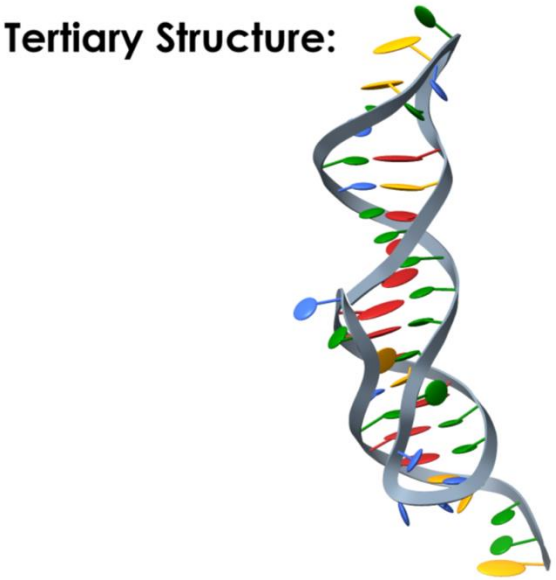

Coding GC 52.27% 1st letter GC 55.72% 2nd letter GC 42.54% 3rd letter GC 58.55%

Sequence/Codon optimization

- mRNA sequence can be optimized in terms of **structural stability** to **enhance protein expression**
- Prioritizing stable structures by identifying sequences that yield folded structures with lower minimum free energy, indicating stable structures, to enhance the overall mRNA stability
- The local structure of the ORF should not be so stable that it resists unfolding and prevents ribosomal complex from move towards the 3′ end of the mRNA



How many possible combinations of codons are there for amino acid sequence Val-Ala-Gly?

Second letter

First letter	U	C	A	G	Third letter
U	UUU, UUC } Phe UUA, UUG } Leu	UCU, UCC, UCA, UCG } Ser	UAU, UAC } Tyr **UAA Stop** **UAG Stop**	UGU, UGC } Cys **UGA Stop** UGG Trp	U C A G
C	CUU, CUC, CUA, CUG } Leu	CCU, CCC, CCA, CCG } Pro	CAU, CAC } His CAA, CAG } Gln	CGU, CGC, CGA, CGG } Arg	U C A G
A	AUU, AUC, AUA } Ile **AUG Met**	ACU, ACC, ACA, ACG } Thr	AAU, AAC } Asn AAA, AAG } Lys	AGU, AGC } Ser AGA, AGG } Arg	U C A G
G	GUU, GUC, GUA, GUG } Val	GCU, GCC, GCA, GCG } Ala	GAU, GAC } Asp GAA, GAG } Glu	GGU, GGC, GGA, GGG } Gly	U C A G

How many possible combinations of codons are there for amino acid sequence Val-Ala-Gly?

- 4 x 4 x 4 = 64 combinations

First letter	Second letter: U	C	A	G	Third letter
U	UUU, UUC Phe UUA, UUG Leu	UCU, UCC, UCA, UCG Ser	UAU, UAC Tyr **UAA Stop** **UAG Stop**	UGU, UGC Cys **UGA Stop** UGG Trp	U C A G
C	CUU, CUC, CUA, CUG Leu	CCU, CCC, CCA, CCG Pro	CAU, CAC His CAA, CAG Gln	CGU, CGC, CGA, CGG Arg	U C A G
A	AUU, AUC, AUA Ile **AUG Met**	ACU, ACC, ACA, ACG Thr	AAU, AAC Asn AAA, AAG Lys	AGU, AGC Ser AGA, AGG Arg	U C A G
G	GUU, GUC, GUA, GUG Val	GCU, GCC, GCA, GCG Ala	GAU, GAC Asp GAA, GAG Glu	GGU, GGC, GGA, GGG Gly	U C A G

How many possible combinations of codons are there for amino acid sequence Val-Ala-Gly?

- 4 x 4 x 4 = 64 combinations

- Spike protein of SARS-CoV-2 has 1273 amino acids
- Approximately 2.4 x 10^{632} combinations!

First letter	Second letter: U	C	A	G	Third letter
U	UUU, UUC Phe; UUA, UUG Leu	UCU, UCC, UCA, UCG Ser	UAU, UAC Tyr; **UAA Stop**; **UAG Stop**	UGU, UGC Cys; **UGA Stop**; UGG Trp	U, C, A, G
C	CUU, CUC, CUA, CUG Leu	CCU, CCC, CCA, CCG Pro	CAU, CAC His; CAA, CAG Gln	CGU, CGC, CGA, CGG Arg	U, C, A, G
A	AUU, AUC, AUA Ile; **AUG Met**	ACU, ACC, ACA, ACG Thr	AAU, AAC Asn; AAA, AAG Lys	AGU, AGC Ser; AGA, AGG Arg	U, C, A, G
G	GUU, GUC, GUA, GUG Val	GCU, GCC, GCA, GCG Ala	GAU, GAC Asp; GAA, GAG Glu	GGU, GGC, GGA, GGG Gly	U, C, A, G

Codon Optimization Tools/Algorithm

IDT INTEGRATED DNA TECHNOLOGIES

JOHNS HOPKINS MEDICINE

Codon Optimization Tool

Read the DECODED article about the Codon Optimization Tool »

Learn more about the Codon Optimization Tool »

Sequence Type: ○ DNA Bases ◉ Amino Acids

RESTRICTION SITE INFORMATION

Product Type: Gene

Organism: Homo sapiens (human)

Single Entry | Bulk Entry

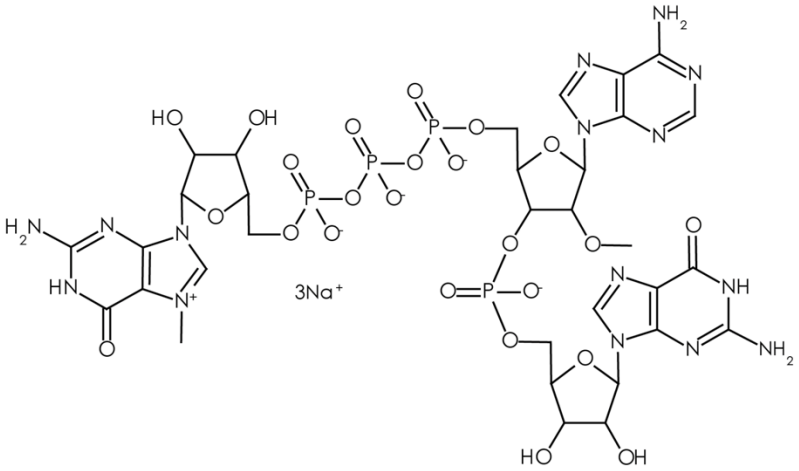
```
IFPKNTCEYIEPVIGHQDMTQWPIWNYWPIAVVIVDFRGGRGPHRVKSKVWFPRCADKPLSWDIFYATRARHQCERHSEYGPASVFYVNHGPSRPLFAFGTLGNWEPSCMKAAGIKFHPVRVKP
CPWNPWCGIFWRANSNGWICYITPLKDCFYEIVMVGCIKYVWCRTCEGHPNGFMHYLINEYHCNTLNPEADRPNKGEDMHQRMFNYKNLHFWRCVVDDHEHIWGPQTTDQGRLRQRVADHHCWI
HWTRSRAMVMALTQNSETVFIKPIMLWMERQNMRAMSGDIVVIGPQCISLWCQTFWRWQQVIIWW
```

# of bases: 313

MAP CODONS ONLY | OPTIMIZE

Cap

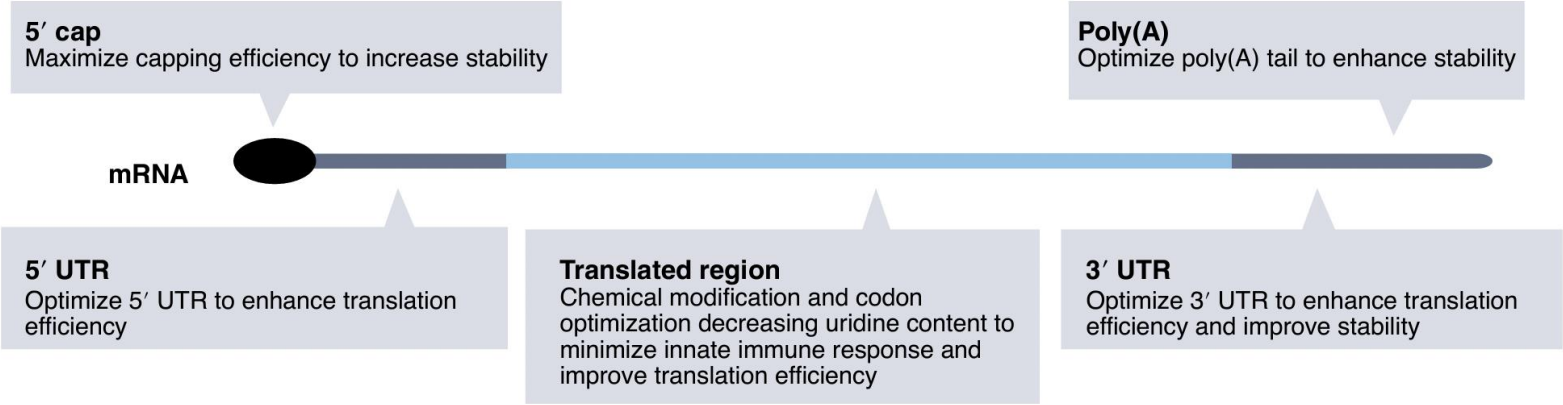
- A modified guanosine nucleotide added to the 5′ end during transcription
- RNA processing and translation factors are recruited to the mRNA cap
- The mRNA cap also protects transcripts from degradation
- Early iterations of cap suffer issues including low mRNA yield and reverse mRNA extension
- Trinucleotide cap analogs solves these issues and can mark mRNA as “self”



Structure of CleanCap Reagent AG N-7113

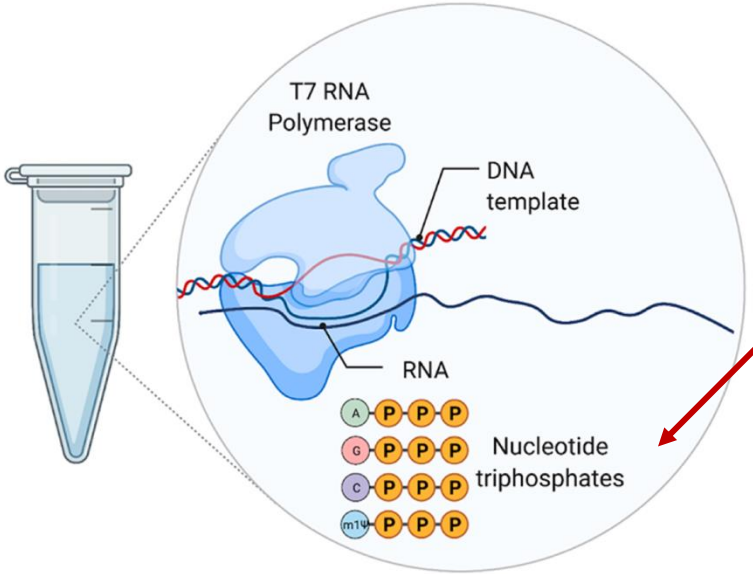
Purpose

- Ultimate purposes
  - Minimize innate immune response
  - Enhance mRNA stability
  - Maximize translation



mRNA manufacturing

Manufacture mRNA – In vitro transcription (IVT)

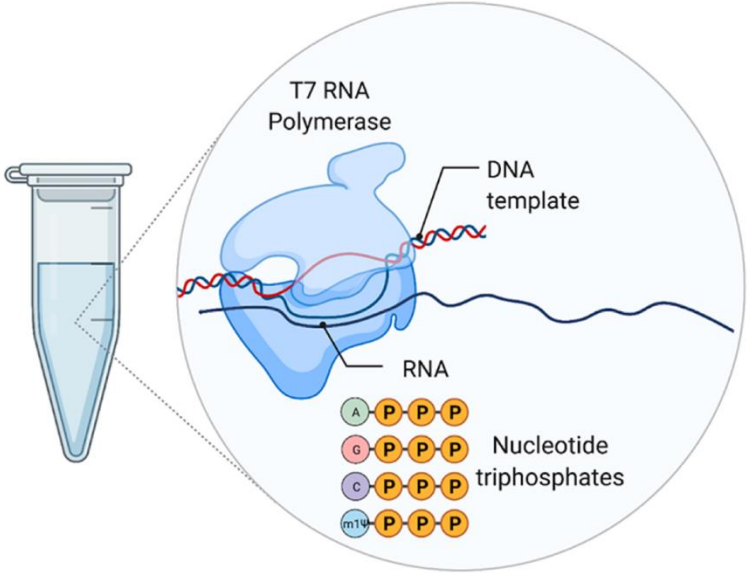


Make sure to include the modified nucleotides, i.e., pseudouridine

IVT allows for template-directed synthesis of RNA molecules from a DNA template in a controlled laboratory environment, outside of living cells

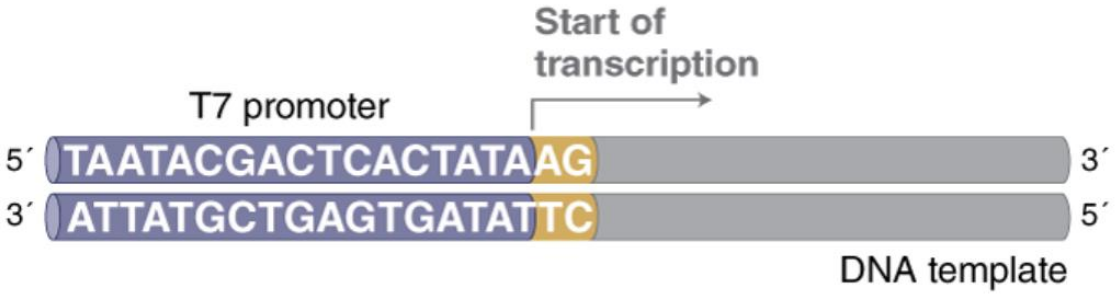
IVT- Capping and Tailing

- Cap and poly(A) tail can both be added co-transcriptionally or post-IVT using specialized enzymes
- Co-transcription capping is done by adding the cap analog as a starting ingredient of the IVT reaction



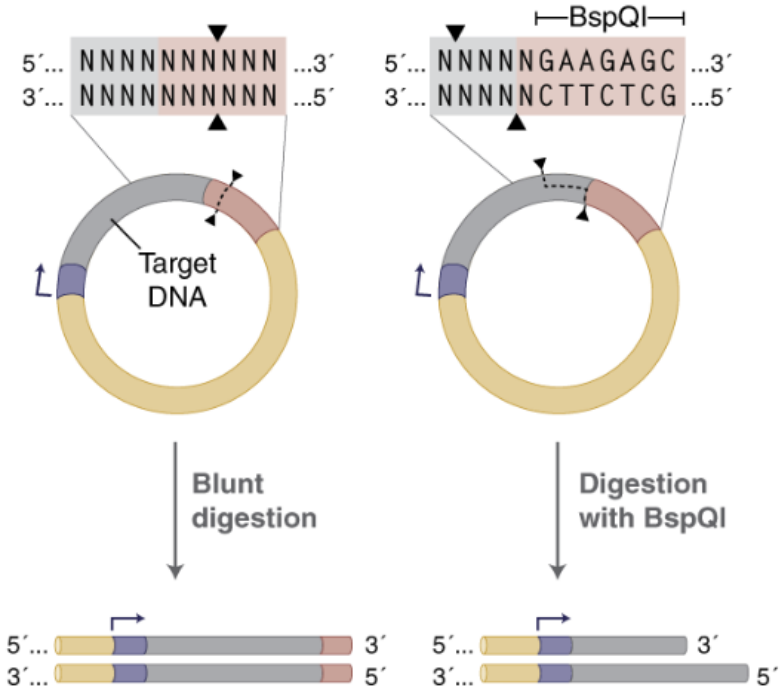
T7 RNA polymerase

- Found in bacteriophage T7
- Has a know promoter sequence
- Can synthesize long RNA transcripts, including messenger RNA (mRNA) using a DNA template
- Capable of producing full-length RNA transcripts with high fidelity

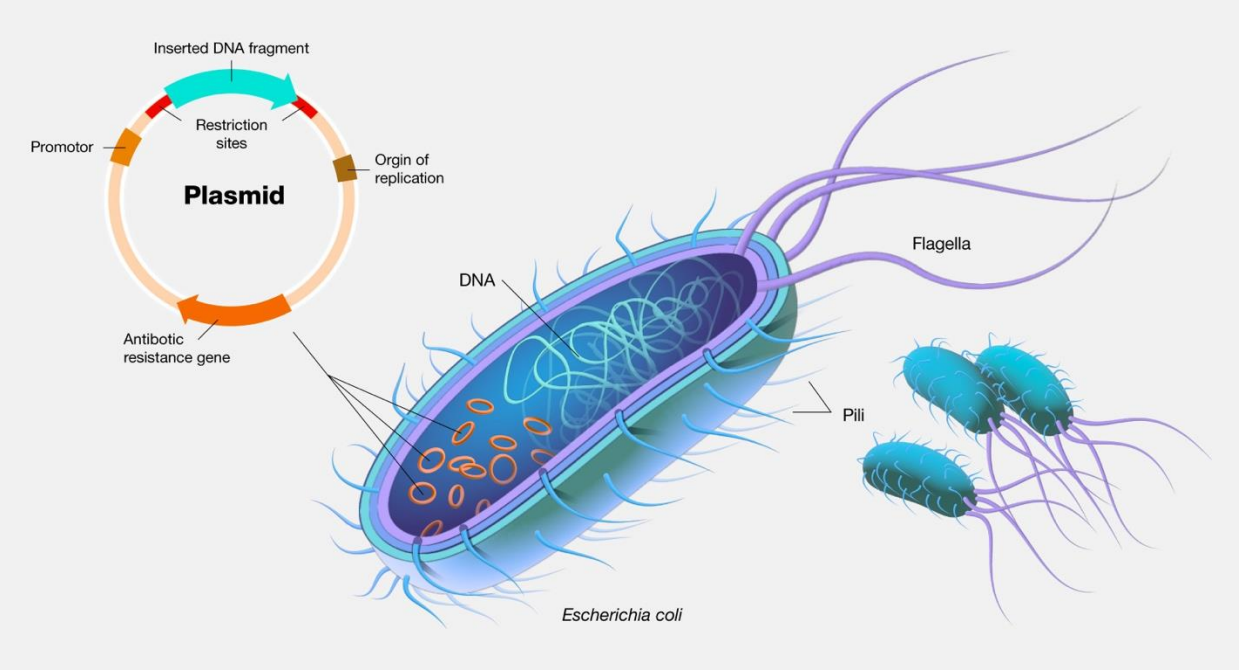


In vitro transcription - DNA template

- Linearized plasmid
- Poly(A) tail sequence can be incorporated in the plasmid
- Linearization is necessary because T7 is an extremely processive enzyme
- T7 will continue to transcribe around a circular template multiple times without disassociating, producing a transcript much longer than the plasmid.
- DNase treatment to remove the DNA template

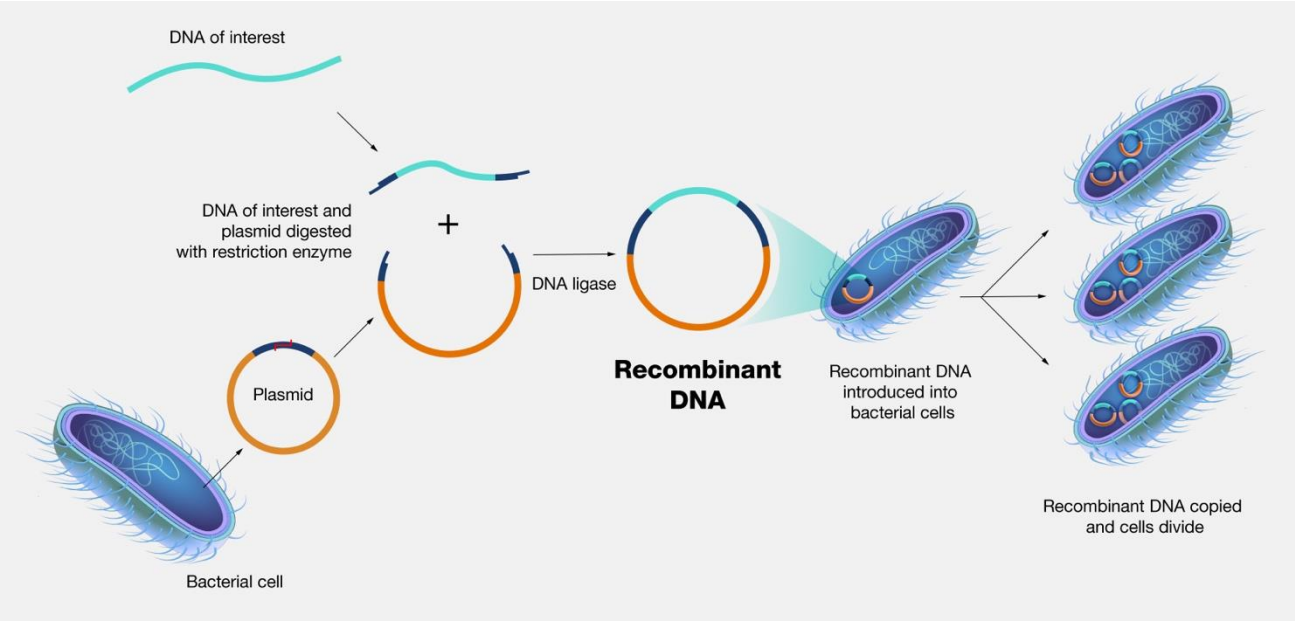


Plasmid



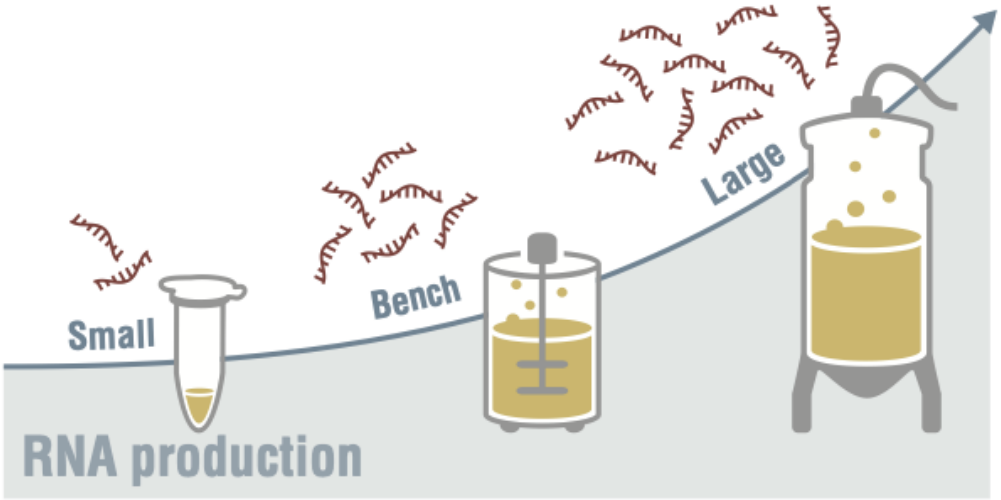
A plasmid is a small circular DNA molecule found in bacteria and some other microscopic organisms

Rapid expansion of DNA templates using bacterial cells



Plasmid containing the DNA template is introduced into bacterial cells. As the bacterial cells divide, the copies of plasmid expand as well

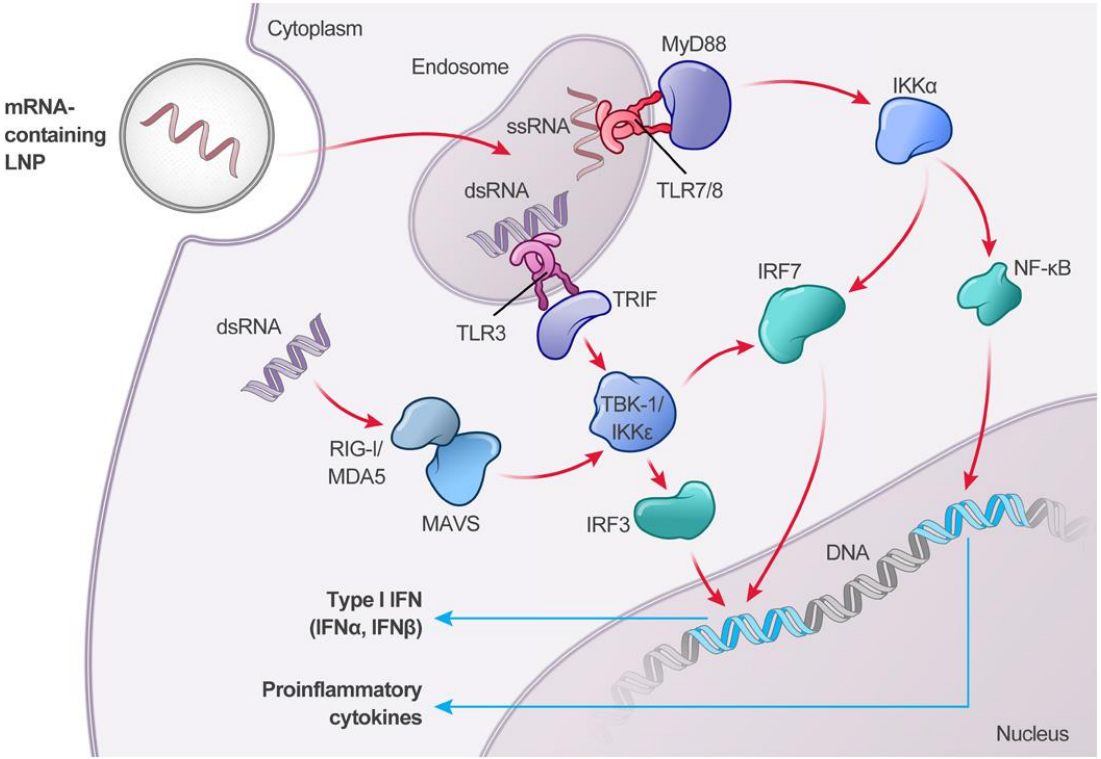
Manufacture mRNA – IVT production scale



IVT allows for synthesis of RNA molecules from short oligonucleotides to those of several kilobases in µg to mg quantities

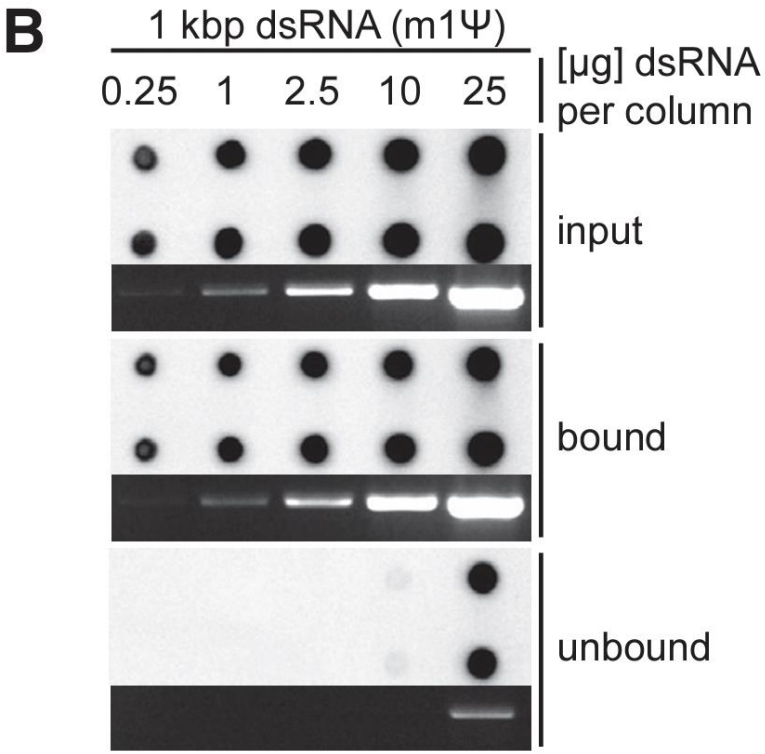
dsRNA as a byproduct of IVT

- Double-stranded RNA (dsRNA) also activates innate immune response
- Sensed in both the endosome and the cytosol
- In endosomes, dsRNA is detected by TLR3, which leads to expression of a large panel of proinflammatory cytokines
- In the cytosol, RIG-I and MDA5 bind to dsRNA and induce expression of proinflammatory cytokines



Purification to remove dsRNA

- Cellulose chromatography
- dsRNA selectively bind to cellulose in an ethanol-containing buffer



Purification to remove dsRNA

- High performance liquid chromatography (HPLC)
- 0.1%-0.5% of the IVT product
- Purified products showed up to 1000-fold higher translation

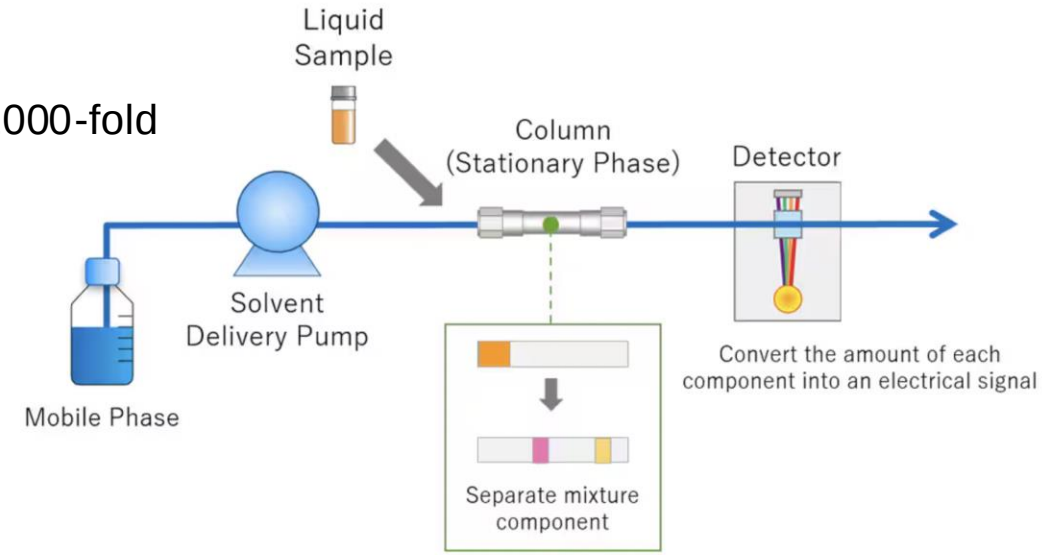
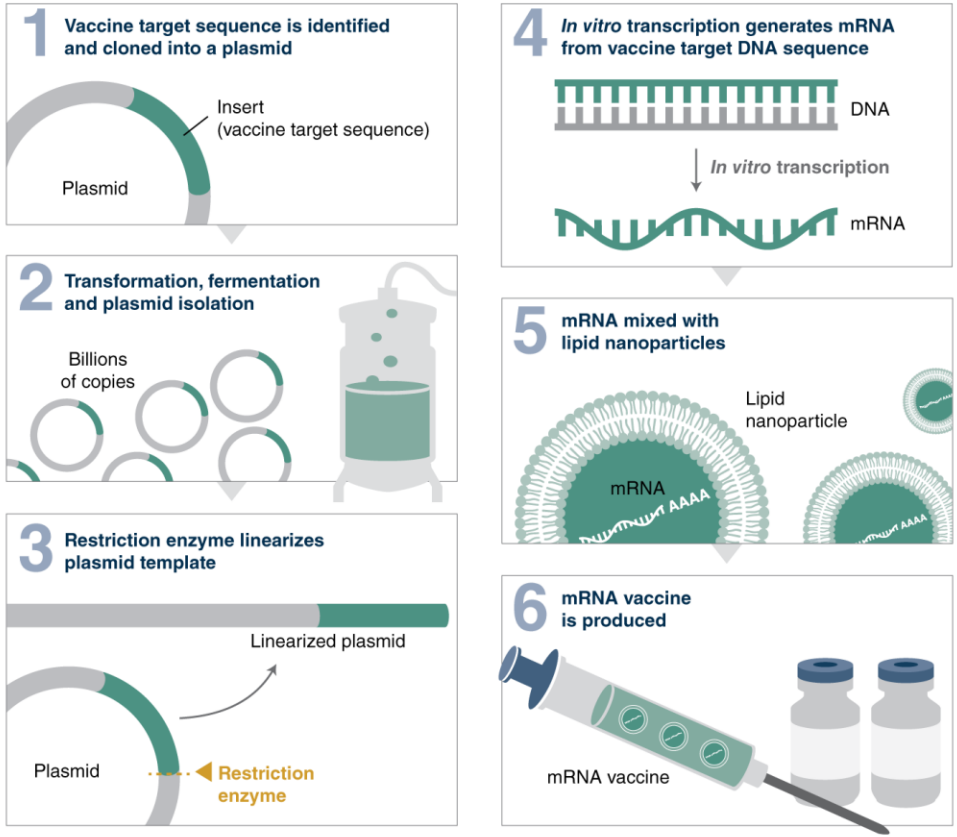


Fig.1 Overview of HPLC

mRNA – from target to product



Created by Elaine Zhelan Chen

mRNA manufacturing

- Rapid production
- Small footprint
- Similar production pipeline for any mRNA sequence
- Suitable for rapid response to emerging infectious disease

Reminders

- Problem set #1
- **Due Tomorrow, Jan 8th, 5PM EST**

- Final presentation team formation
- **Before class tomorrow**

Attendance